

Dimensionality Reduction and Discriminant Analysis

PCA, FDA, LDA and QDA on MNIST

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1 Overview of the Pipeline

The complete pipeline is implemented in a single Python file. The workflow is as follows:

1. Load MNIST dataset using sklearn
2. Select digits 0, 1 and 2 (100 samples each)
3. Apply PCA for dimensionality reduction
4. Apply Fisher Discriminant Analysis (FDA)
5. Train LDA and QDA classifiers
6. Compute training and test accuracy
7. Visualize 2D projections

2 Dataset Preparation

Only digits 0, 1, and 2 are selected. Exactly 100 samples per digit are randomly chosen for both training and test sets. All pixel values are normalized to $[0, 1]$.

```
Xtr, ytr, Xte, yte = load_mnist_subset()
```

Each image is represented as a 784-dimensional feature vector.

3 Principal Component Analysis (PCA)

PCA reduces dimensionality by projecting data onto directions of maximum variance.

3.1 Covariance Matrix

$$S = \frac{1}{N-1}(X - \mu)(X - \mu)^T$$

```
mu = np.mean(X, axis=1, keepdims=True)
Xc = X - mu
S = (Xc @ Xc.T) / (X.shape[1] - 1)
```

3.2 Eigen Decomposition

Eigenvectors corresponding to largest eigenvalues are selected.

```
eigvals, eigvecs = np.linalg.eigh(S)
idx = np.argsort(eigvals)[::-1]
eigvecs = eigvecs[:, idx]
```

3.3 PCA Reconstruction

To evaluate how much information is preserved after dimensionality reduction, the original images are reconstructed from their PCA representation.

After computing the principal components, the mean-centered data is projected onto the reduced subspace:

$$Y = U_p^T(X - \mu)$$

where U_p contains the eigenvectors corresponding to the largest eigenvalues.

The approximate reconstruction of the original data is obtained by projecting the reduced representation back to the original space:

$$\hat{X} = U_p Y + \mu$$

Here μ is the mean vector that was subtracted during preprocessing and added back after reconstruction.

Five samples from the training set were reconstructed using the principal components that retain 75% of the total variance.

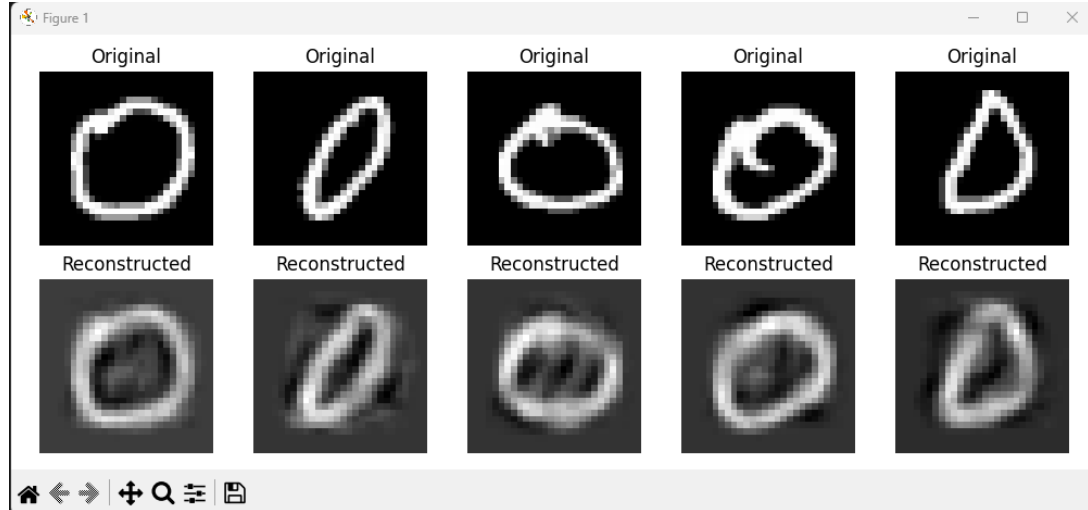


Figure 1: Original images (top row) and reconstructed images using PCA (bottom row).

The quality of reconstruction is measured using the Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2$$

where x_i represents the original image and $\hat{x}_i$ represents the reconstructed image. For the reconstructed samples, the observed reconstruction error is:

$$\text{MSE} = 0.02379$$

4 Fisher Discriminant Analysis (FDA)

FDA is a supervised dimensionality reduction technique.

It maximizes the ratio:

$$J(W) = \frac{W^T S_B W}{W^T S_W W}$$

4.1 Between-Class Scatter

$$S_B = \sum_c N_c (\mu_c - \mu)(\mu_c - \mu)^T$$

4.2 Within-Class Scatter

$$S_W = \sum_c \sum_{x \in c} (x - \mu_c)(x - \mu_c)^T$$

```
mat = np.linalg.pinv(SW) @ SB
eigvals, eigvecs = np.linalg.eigh(mat)
```

For 3 classes, FDA provides at most $C - 1 = 2$ projection dimensions.

5 Classification using LDA and QDA

5.1 LDA

$$g_c(x) = x^T \Sigma^{-1} \mu_c - \frac{1}{2} \mu_c^T \Sigma^{-1} \mu_c + \log \pi_c$$

5.2 QDA

$$g_c(x) = -\frac{1}{2} \log |\Sigma_c| - \frac{1}{2} (x - \mu_c)^T \Sigma_c^{-1} (x - \mu_c) + \log \pi_c$$

Classification is done by selecting the class with maximum discriminant score.

6 Accuracy Results

Method	Train Accuracy	Test Accuracy
Raw LDA	1.0000	0.9733
Raw QDA	1.0000	0.9867
PCA (75%) + LDA	0.9733	0.9633
PCA (75%) + QDA	1.0000	0.9867
PCA (90%) + LDA	0.9900	0.9667
PCA (90%) + QDA	1.0000	0.9867
First 2 PC + LDA	0.9100	0.9033
FDA + LDA	1.0000	0.9267
FDA + QDA	1.0000	0.9267

7 PCA Visualization

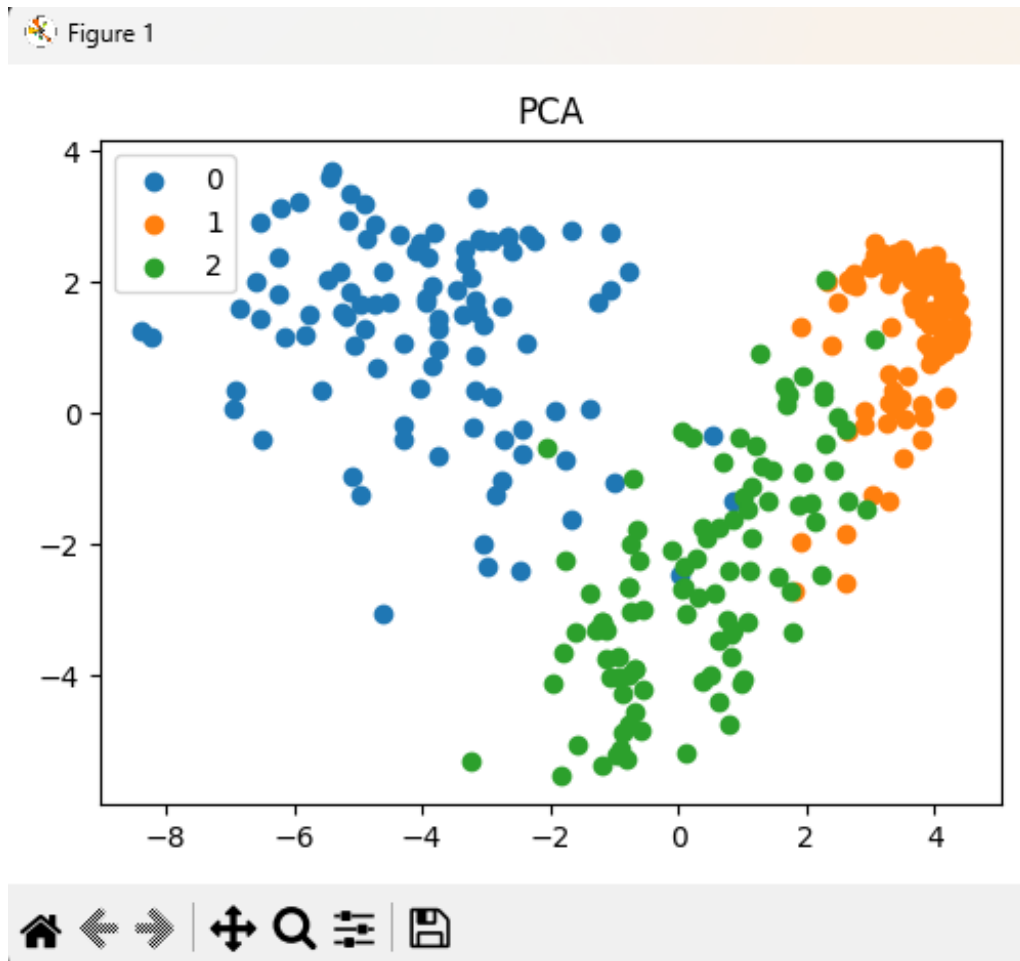


Figure 2: Projection of training samples onto first two principal components

8 FDA Visualization

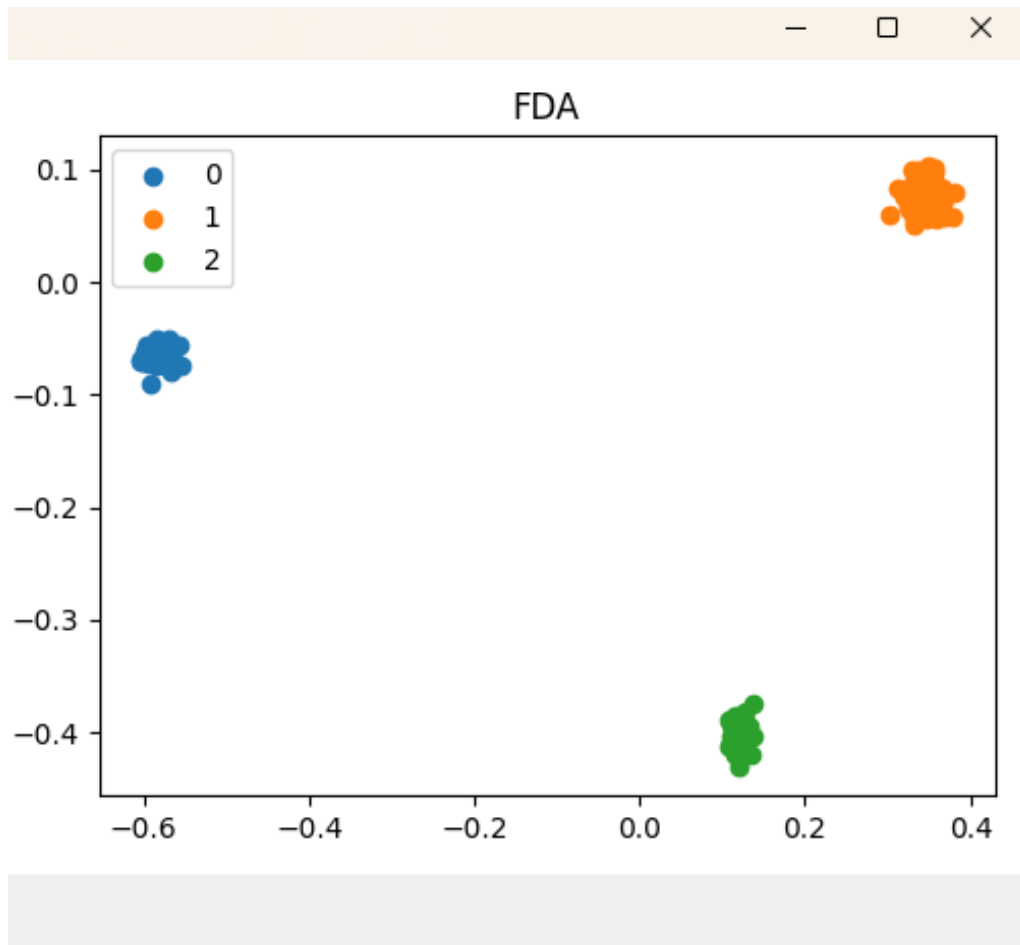


Figure 3: Projection of training samples onto two Fisher discriminant directions

9 Observations

- PCA maximizes variance but does not use class labels.
- FDA improves class separation in low dimensions.
- QDA consistently performs better due to class-specific covariance modeling.
- Using only two principal components significantly reduces accuracy.